Final Project

Introduction to Computational Social Science

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Necessary Background

This is your final project for the "Introduction to Computational Social Science" course. Over the next few weeks, you will work as part of a team of computational social scientists, tackling an important social science problem of your choice. Together with your teammates, you will formulate a research question and develop a suitable research strategy, find or collect the necessary data, conduct your analysis, prepare a preliminary team presentation, and write a final research paper that summarises your findings and process.

This document contains all the information you need to effectively organise your work and ensure a successful research project. Embrace this opportunity to collaborate, learn, and even make meaningful contributions to the field. Let's make the most of this experience!

What we are supposed to do?

In this final project, you have **multiple tasks**:

1. Formulate a **research question** that is **relevant** to the field of Computational Social Science.
2. Develop a **research strategy** that effectively addresses your research question.
3. Identify the necessary **data** and conduct your analysis using Python or R.
4. Present the **results** of your project during the final sessions of the course.
5. Prepare a **final research paper** (approximately 5,000 words) summarising your analysis and findings, and submit it in PDF format.

Class Presentation

Class Presentation is organised as a simulation of an academic conference. Your presentation should consist of approximately 10 slides, designed to support a talk lasting around 10 to 15 minutes, followed by an additional 5 minutes for questions. Focus on telling a compelling story about your research journey. Clearly articulate the significance of your results and try to convince the audience of their importance. This is an opportunity to showcase your findings and engage with your peers, so make sure to highlight the key elements of your project.

Final Research Paper

Your final research paper should be a written report of around 12 pages (around 5,000 words), including 2 to 6 figures and/or tables. Consider this as a mini-paper, structured in the following manner:

- **Introduction**: Clearly state and motivate your research question, providing context and relevance to the field of Computational Social Science, and include some relevant papers.

- **Data Description**: Describe the data you will use to explore your research question, including its source, characteristics, preprocessing steps taken, and relevant data analysis.
- **Results**: Present your findings and use figures or tables to enhance clarity. Explain what your results mean in relation to your research question.
- **Discussion**: Discuss how your results answer your research question, considering implications and potential limitations of your study.

NB: In research, it is normal to encounter results that challenge expectations or seem unclear. Your goal in this course is to explore social science issues using computational methods — not to produce groundbreaking results. Focus on engaging with the process and applying the tools appropriately.

Data

When conducting your research, you can use your own data or access various datasets available online. Below are some reliable sources where you can find datasets relevant to your project:

- SocioPatterns datasets (social and animal networks): sociopatterns.org
- Stanford Large Network Dataset Collection (different network data): snap.stanford.edu
- Awesome Public Datasets (many datasets on different topics): github.com
- Kdnuggets (many datasets on different topics): kdnuggets.com

Can I use ChatGPT or other AI tools?

Since this course is not focused on programming, you **are welcome** to use AI tools if you find yourself struggling to come up with the right **coding** solutions. However, please be transparent about this and acknowledge your use of AI in the relevant parts of your code. This will not affect your grade, but it will demonstrate your thoughtfulness and respect for the course instructors.

Grading

Overall, your grade for the Final Project will be based on a total of **100 points**, distributed across the following two categories:

Class Presentation - 50 points

Table 1: Rubric Assessment Criteria: Class Presentation

Criterion	Fully Completed (10 points)	Partially Completed (6 points)	Needs Improvement (3 points)
Relevance for Computational Social Science	Strongly connects the project to Computational Social Science, highlighting its significance.	Some relevance, but lacks clarity or depth.	Unclear connection to Computational Social Science.
Clarity of Presentation	Clear and coherent delivery, easy for the audience to follow.	Somewhat clear, but may confuse the audience.	Unclear or disorganised, hindering understanding.
Use of Visual Aids	Effective use of visual aids that enhance understanding.	Visual aids present but lack clarity or support.	Poorly designed or absent visual aids.
Content and Analysis	Well-researched content with insightful analysis.	Adequate content, but lacks depth in analysis.	Superficial or poorly researched content.
Handling Questions	Confidently and thoughtfully responds to questions.	Answers adequately, but lacks depth or confidence.	Struggles to respond, showing limited understanding.

In special cases, points may be awarded in increments that fall between the established steps to better reflect the quality of your work.

Final Research Paper - 50 points

NB: Submit the Final Research Paper in PDF format along with the **code** used for data analysis.

Table 2: Rubric Assessment Criteria: Final Research Paper

Criterion	Fully Completed (10 points)	Partially Completed (6 points)	Needs Improvement (3 points)
Coding	Code is clean, well-organised, fully functional, and well-documented. The use of AI is not being abused.	Code is mostly well-structured, with clear functions; comments and documentation are present but may be sparse.	Basic functionality is implemented, but the code may lack structure or proper documentation.
Motivation and Literature	Effectively motivates the research question, thoroughly reviewing relevant literature and situating the research within the existing body of knowledge.	Provides some motivation and literature review, but lacks depth or misses key connections.	Insufficient motivation or literature review, lacking relevance to the field.
Methodology	Clearly describes the research design and methods for data collection and analysis suitable for Computational Social Science.	Methodology is mentioned but lacks detail or clarity in the context of Computational Social Science.	Inadequate description of methods, hindering reproducibility.
Results and Discussion	Thorough presentation and interpretation of results, with clear connections to the research question and implications for Computational Social Science.	Results are presented but analysis lacks depth or clarity regarding implications for the field.	Results are poorly presented or inadequately discussed.
Writing Quality	Well-structured, clear, and coherent writing with minimal errors.	Writing is generally clear but contains some errors or lacks cohesion.	Writing is unclear or disorganised, with frequent errors.

In special cases, points may be awarded in increments that fall between the established steps to better reflect the quality of your work.